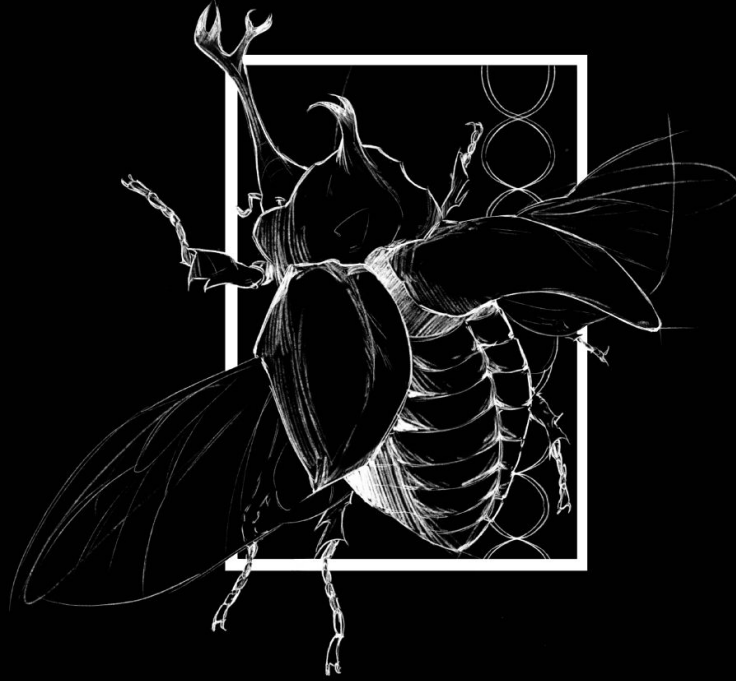




Background

Across the Gekkonidae lineage, the evolutionary trait of acoustic vocalization is a vital survival mechanism in territorial and reproductive communication. **This study analyzes the role of geography in influencing vocal signaling, specifically to determine if the loss of vocalization is linked to geographic isolation and loss of predation.** Current research suggests that geographic isolation on islands triggers a loss of behavioral traits where reduced predation leads to relaxed selection. In using a comparative analysis approach, we can asses the gain and loss of acoustic vocalization among shared evolutionary histroy. In this study, the family, gekkonidae, is studied on their geographic location and vocal status. In looking at this relationship, I will determine if island occupancy effects vocalization across the gekkonidae lineage.

Study Design & AI

For this study, a phylogenetic approach was used to assess the relationship between geographic isolation and acoustic vocal status. Data for 66 species across the Gekonidae lineage were found and their traits were mapped across a phylogeny tree constructed using AI. Generative AI was used in this study to create and troubleshoot R scripts and conduct internet wide data research. Furthermore, AI was employed to conduct a phylogenetic comparison method, the Pagels Test, to analyze the correlated evolution between the two traits. Overall, the incorporation of AI into this study helped in performing a research gap analysis in gekkonidae vocalization and construct phylogenies and phylogenetic comparison models.

Analytical Approach

While the gekkonidae family is a large, diverse lineage, many species faced the problem of lacking sufficient data. Using the 66 species, the relationship between geographic isolation and acoustic vocalization was evaluated using Pagels Test as the primary phylogenetic comparison method. This method tests the codependence of these traits and identifies if the loss of vocalization across the phylogenetic tree is linked to island isolation. Furthermore, a phylogenetic logistic regression was conducted to determine the strength in the relationship between the two traits.

Variable	Estimate	StdErr	z.value	p.value
(Intercept)	-1.2473	1.0624	-1.1741	0.2404
insular_bin	-0.0002	0.4905	-0.0005	0.9996

Results

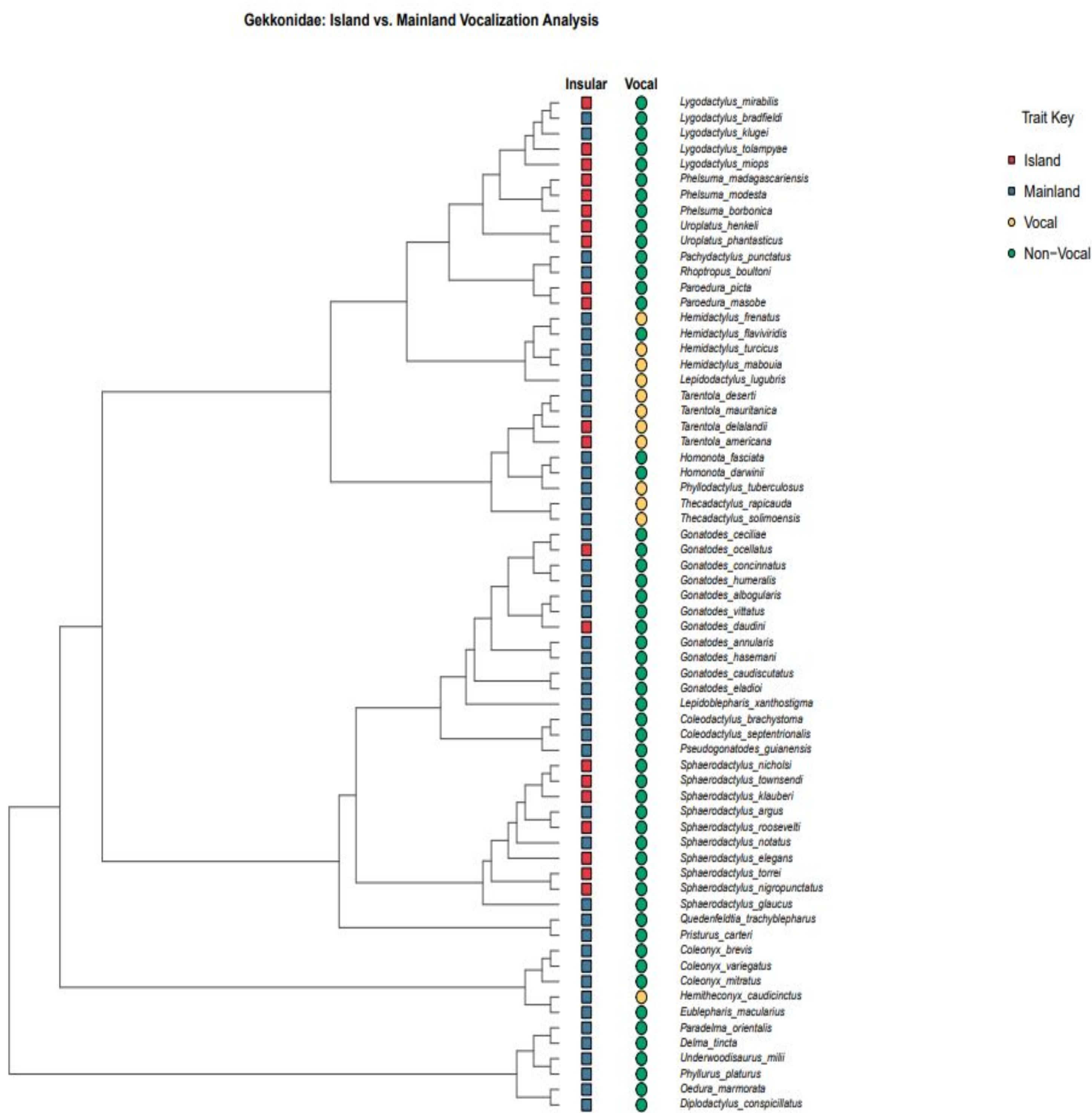


Figure 1: A phylogeny of 66 species across the gekkonidae lineage are mapped with their vocal status and insularity. This phylogeny shows complex distrution of these traits across shared evolutionary history.

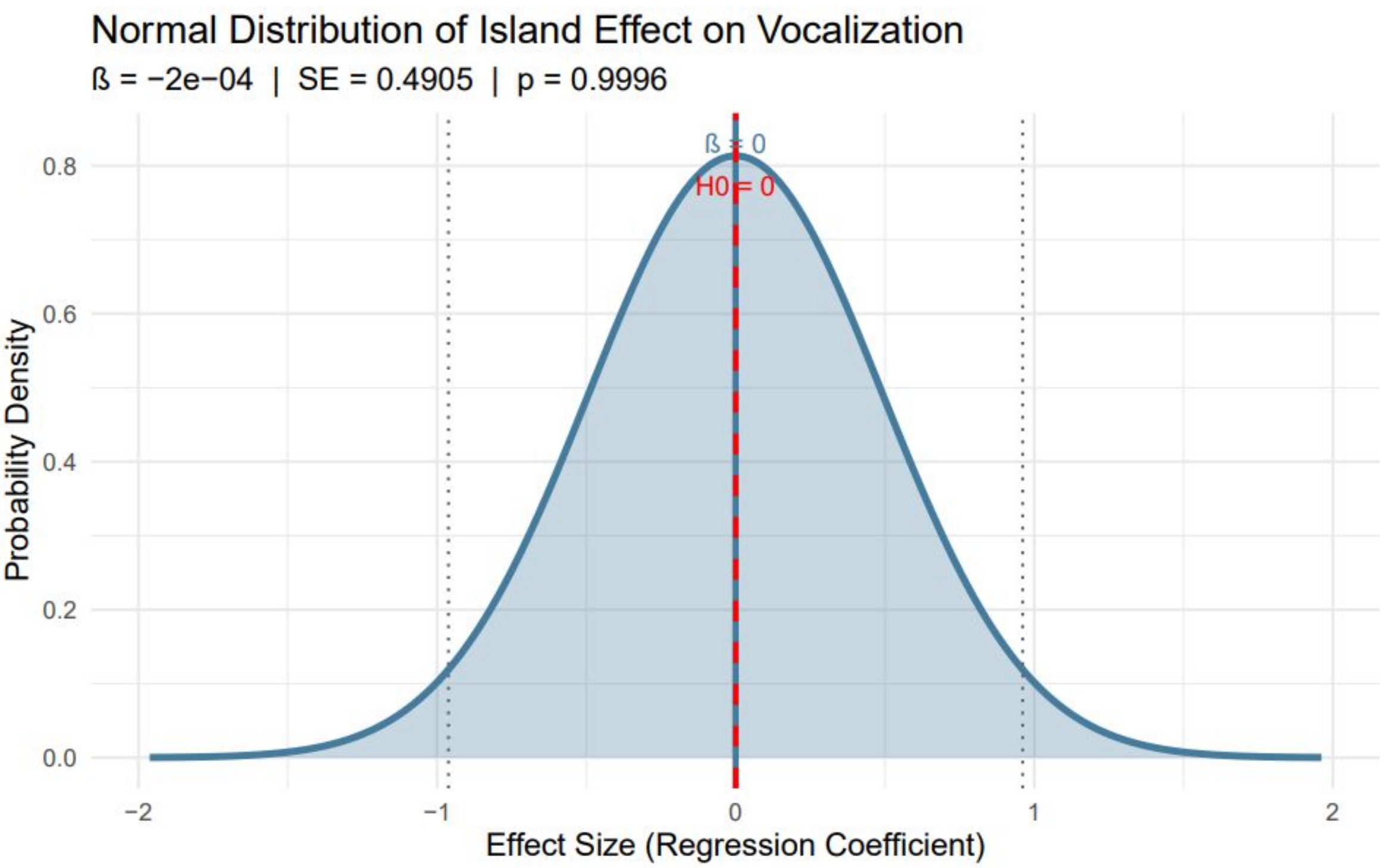


Figure 2: A normal distribution of the phylogenetic logistic regression values. The peak around 0 indicates the failure to reject the null hypothesis and the robustness of the vocalization trait in island isolation.

Discussion

In using the phylogenetic comparison analysis methods, the p value obtained from the pagels tests strongly concludes that the there is no statistically significant relationship between the two variables. The obtained p value, 0.9996, indicates that the the evolution of acoustic vocalization across the gekkonidae lineage is robust and stable even after long term island exposure .

While island isolation and loss of predation predicts the loss of complex behavioral traits, the results obtained in this study show that geckos retain their vocal status regardless of the of their environment. With island isolation having no effect of geckos ability to vocalizae, other factors can possibly be attributed to this traits simplification.

By uitlizing the phylogenetic approach in this study, we can confirm the stability of the acoustic vocalization trait is consistent across the gekkonidae lineage. Furthermore, this study demonstrates the limitations of reducing complex behavioral traits down to binary data. Binary coding may mistakenly confuse simple distress chirps as complex calls. It is essential to acknowledge the large data gaps found across the lineage and the need for continued research to develop a better understanding of how isolation effects vocalization across the gekkonidae family. Furthermore, future incorporation of continuous data, like frequency, may show effects on the quality of vocalization, but more research needs to be conducted before a relationship is established.

Conclusions & Acknowledgements

- Given that the p value is greater than 0.05, I fail to reject my null hypothesis. It is concluded that vocalization evolves independently of geographic isolation.
- With geographic isolation increasing the relaxation of behavioral traits, geckos vocal status is robust to long term island exposure
- With the large gaps present in gecko vocalization data, incorporation of continuous traits, like frequency, can develop the understanding of the relationship between these traits further.